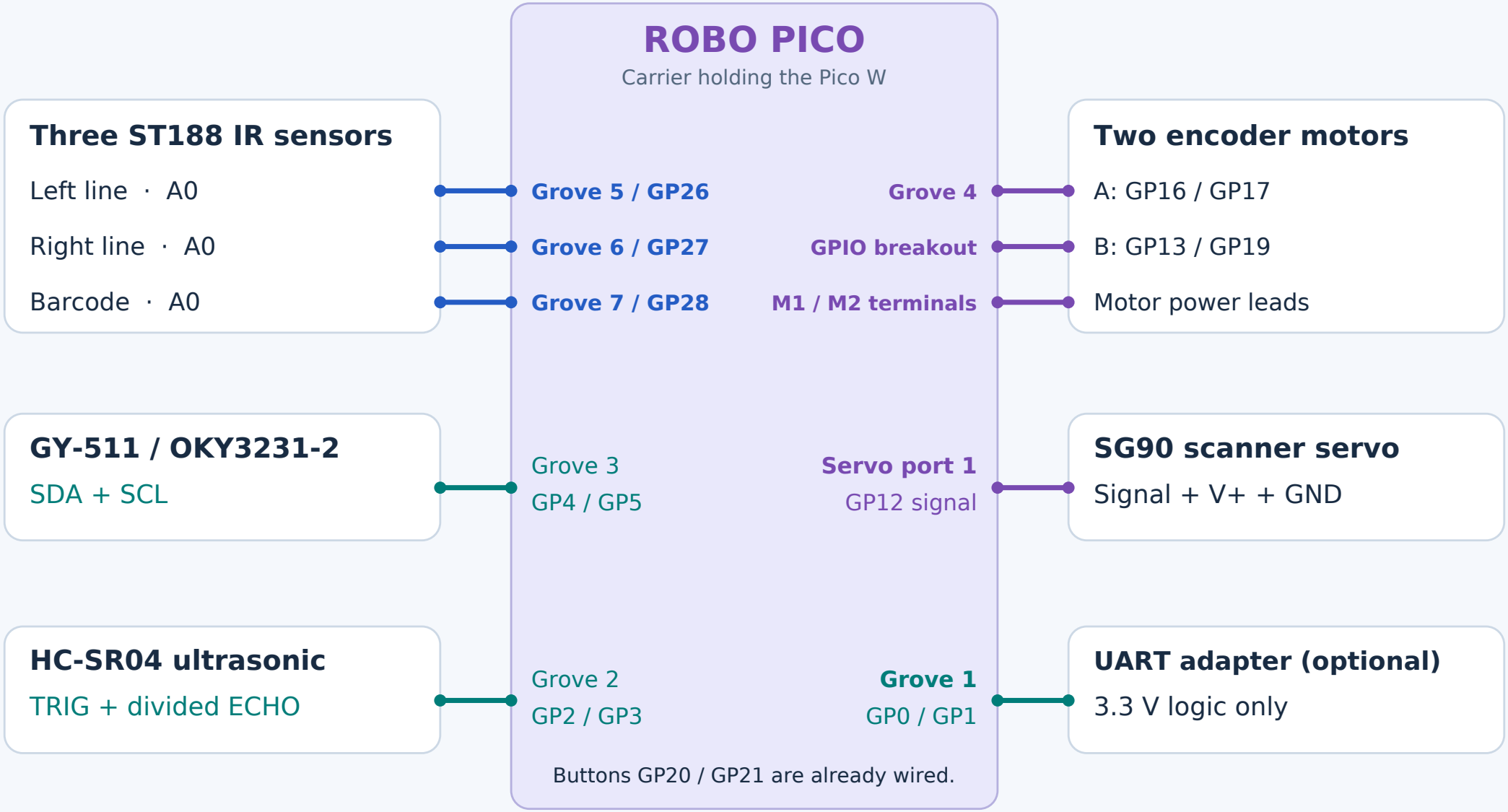
01 / What connects where

Start here. Each group has its own detailed sheet.

Signal overview only. Follow sheet 07 for power and common ground.

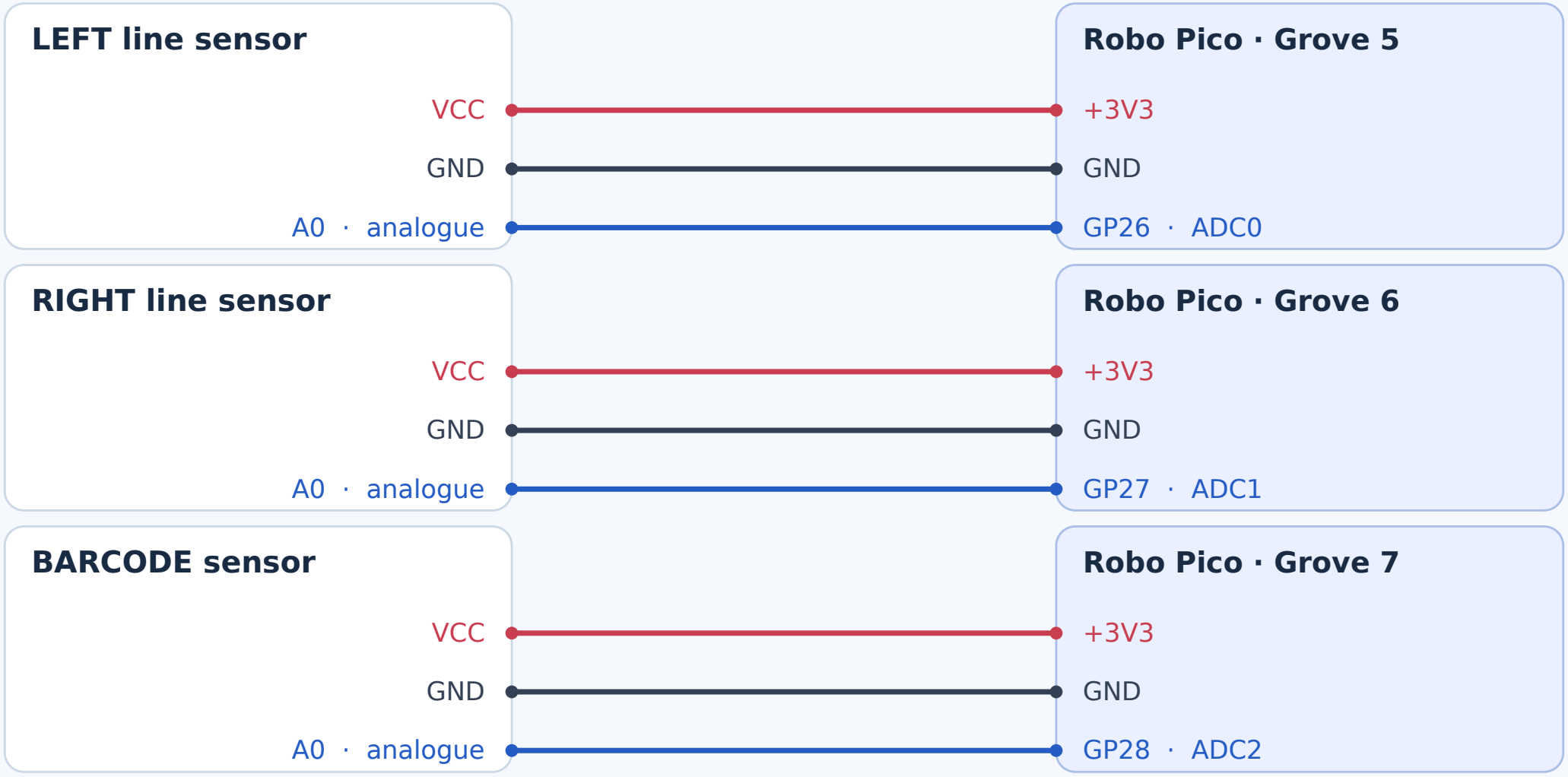


Follow printed pin labels. Contacts are grouped by function, not physical position.

ROBO-PICO datasheet pp. 3–4 + PicoCar slides + current board.c assignments

02 / ST188 sensors: three wires each

Use A0 (analogue). Leave D0 (digital) disconnected.



Disconnect unused signal wires; use a repinned/breakout cable.

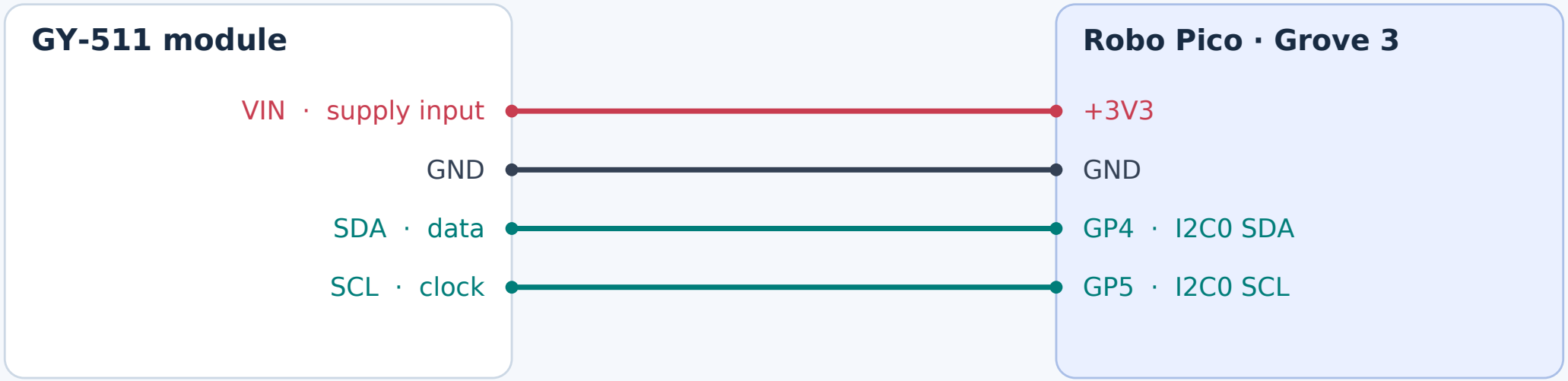
Grove 5: no GP6 wire. Grove 6: no GP26 wire. Grove 7: no GP7 wire. Open GP28 VBAT jumper.

Follow printed pin labels. Contacts are grouped by function, not physical position.

PicoCar slide 4 labels + ROBO-PICO Grove mapping + IR manual supply limits

03 / GY-511 / OKY3231-2 IMU

LSM303DLHC accelerometer · I2C address 0x19



Leave these module pins unconnected:

3.3 pad, INT1, INT2 and DRDY. VIN and the separate 3.3 pad are not interchangeable labels.

Use this module's VIN input with the 3.3 V arrangement shown.

Verify operation and pull-up voltage. Do not feed 5 V onto SDA or SCL; do not attach two supplies.

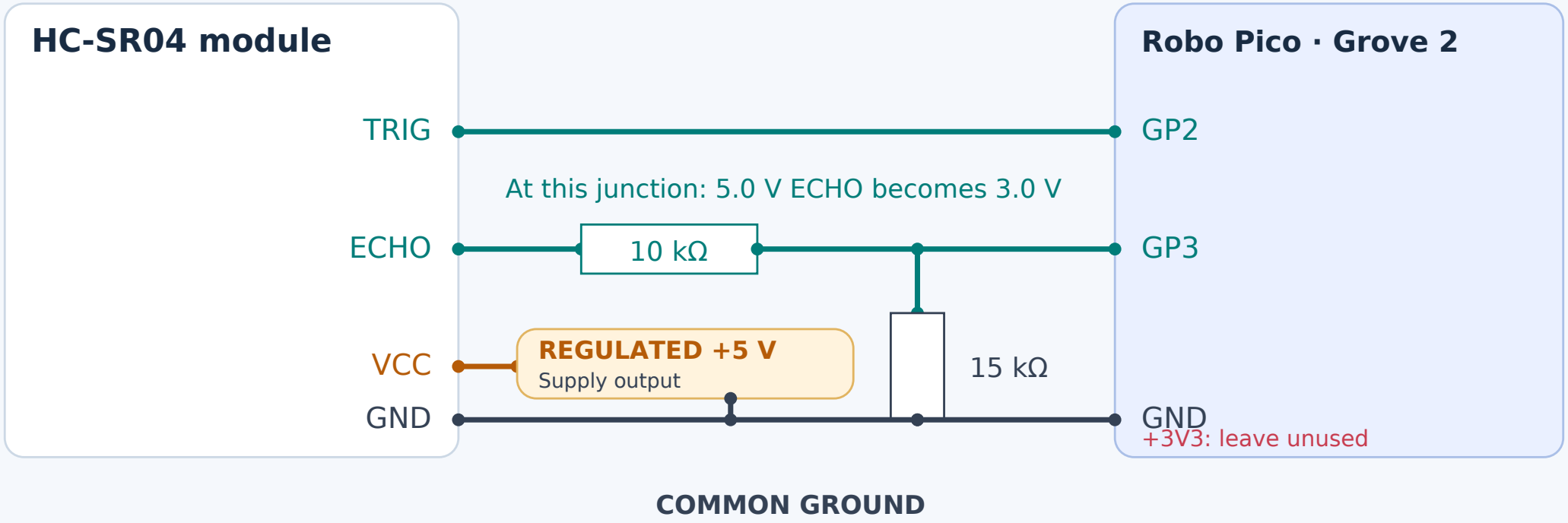
The module has an accelerometer and magnetometer, not a gyroscope.
Use the printed SDA/SCL/VIN/GND labels; do not count positions from the drawing.

Follow printed pin labels. Contacts are grouped by function, not physical position.

OKY3231-2 module sheet: 3–5 V input + PicoCar slide 5 labels + LSM303DLHC datasheet

04 / HC-SR04: ECHO needs two resistors

HC-SR04 supply = regulated nominal 5 V. Pico signal level = 3.3 V.



Supply GND, sensor GND and Robo Pico GND are connected together.

ECHO → 10 kΩ → GP3 junction; 15 kΩ from that same junction to GND.

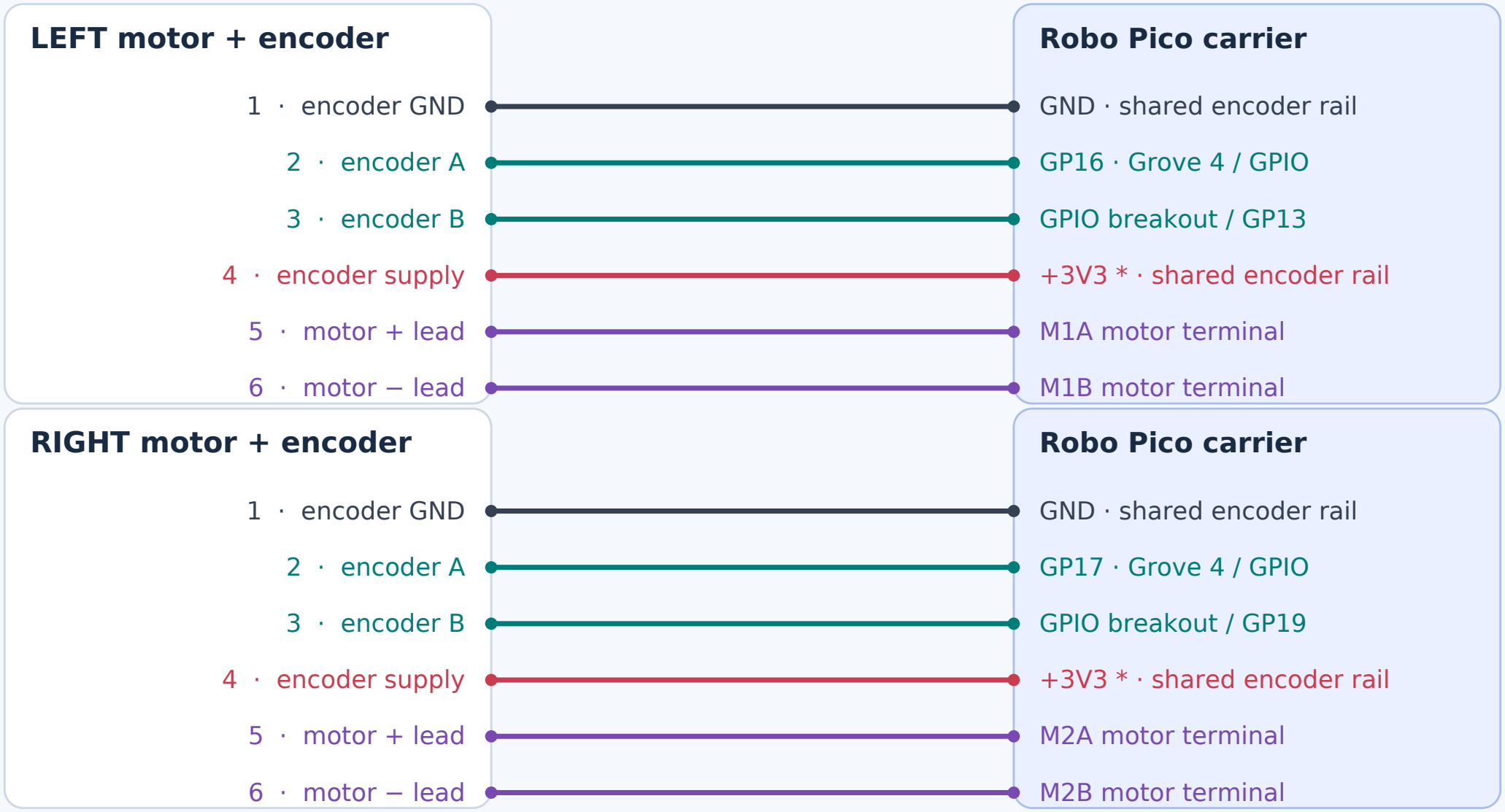
Do not connect ECHO directly to GP3. Servo V+ is only an alternative if verified as a suitable 5 V supply.

Follow printed pin labels. Contacts are grouped by function, not physical position.

HC-SR04 datasheet: 5 V supply; RP2040/Robo Pico input limits; divider 15 / (10 + 15)

05 / Encoder motors: six wires per motor

Motor-connector numbers are from the PicoCar diagram. Confirm its orientation on your unit.



* 3.3 V encoder supply is proposed in the slides; verify the actual encoder's voltage/current rating.

Follow printed pin labels. Contacts are grouped by function, not physical position.

PicoCar slide 2: PH2.0-6P functions; owner marking KH2.0-6P differs; ROBO-PICO M1/M2 mapping

05B / One Grove socket, or individual jumpers

GPIO breakout = the GP-labelled female header sockets on the Robo Pico carrier.



A Grove breakout cable sends the two different signal contacts to different motors.



With individual jumper wires, you can leave Grove 4 EMPTY. No firmware change.

Both encoder pin 1 wires share GND. Both pin 4 wires need their rated encoder supply.

Distribute encoder power/GND through a small breadboard or terminal block; use 3.3 V only if rated.
Keep motor power (pins 5/6) on M1/M2. Never join left A and right A together.

Follow printed pin labels. Contacts are grouped by function, not physical position.

Grove 4 exposes GP16/GP17; the carrier GPIO headers expose those same electrical signals.

06 / SG90 servo + optional console

The START/RESET and E-STOP buttons are already connected on the carrier.



Servo colours match the slide; verify your unit before inserting its plug.

Servo port 2 / GP13 is reserved for encoder B. UART: 115200 baud, 3.3 V logic; leave adapter VCC off.

ONBOARD GP20 BUTTON

START / RESET · no jumper needed

ONBOARD GP21 BUTTON

E-STOP · no jumper needed

Follow printed pin labels. Contacts are grouped by function, not physical position.

PicoCar slide 3 lead identification + ROBO-PICO servo/UART/button mapping

07 / Power: keep these supplies distinct

All sensor, controller and external-supply grounds must share common GND.

GROVE +3V3

ST188 VCC × 3 (3.3 V-compatible boards)

GY-511 VIN (module input rated 3–5 V)

Encoder pin 4 × 2, only if rated for 3.3 V

Total Grove 3.3 V output budget: 300 mA

REGULATED NOMINAL 5 V

HC-SR04 VCC

Use an external regulated supply, OR servo V+ only after measuring a suitable voltage under motor load.

Not Grove +3V3. Keep the ECHO divider installed.

SERVO V+ / MOTOR DRIVER

Servo V+ follows the board's power source.

It is not a fixed 5 V regulator output.

SG90 must tolerate the measured servo supply.

Motor leads go to M1/M2, never directly to GPIO.

ROBO PICO BOARD INPUT

Datasheet input range: 3.6–6 V.

Use its intended power connector and correct polarity.

M1 / M2: 1 A continuous per channel, 1.5 A <5 s peak.

Do not tie an external 5 V output to Grove +3V3.



Follow printed pin labels. Contacts are grouped by function, not physical position.

ROBO-PICO power limits: input 3.6–6 V, Grove 3V3 total 300 mA, motor 1 A continuous / channel